



PROJECT REPORT

House Price Prediction Using Linear Regression

Artificial Intelligence & Machine Learning – Task 1

Submitted by:

Name: Asma

Intern ID: MT7065

Role: ML Intern

Internship Track: Artificial Intelligence &
Machine Learning (AI/ML)

Submitted to:

Organization: Maincrafts Technology

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Introduction

This project builds a Supervised Machine Learning pipeline using **Linear Regression** to predict continuous California house values (HouseValue). The workflow covers the entire ML lifecycle: data loading, exploratory data analysis (EDA), preprocessing, train-test splitting, feature scaling, model training, evaluation, diagnostic visualization, and model persistence.

Dataset Description

The benchmark **California Housing Dataset** contains **20,640 records** representing block groups from the 1990 U.S. Census, consisting of **8 numerical features** and **1 continuous target variable**:

- **Features:** MedInc (Median Income), HouseAge, AveRooms (Avg Rooms), AveBedrms (Avg Bedrooms), Population, AveOccup (Avg Occupancy), Latitude, Longitude.
- **Target:** HouseValue (Median house value scaled in hundreds of thousands of dollars, e.g., \$2.5 = \ \$250,000\$).

1. Import Libraries

- **Data Core:** pandas, numpy for manipulation and array processing.
- **Graphics:** matplotlib.pyplot, seaborn for histograms, heatmaps, and diagnostic plots.
- **ML Pipeline:** sklearn for dataset sourcing, train_test_split, StandardScaler, LinearRegression, and metrics computation (mean_absolute_error, mean_squared_error, r2_score).
- **Persistence:** pickle for model serialization.

2. Load the Dataset

Sourced via fetch_california_housing(as_frame=True). The feature matrix and target vector were mapped into a single unified Pandas DataFrame, with the target column explicitly labeled as HouseValue.

3. Exploratory Data Analysis

- **Missing Values:** Ran df.isnull().sum(), confirming **0 missing values** across all 20,640 records. No imputation was required. The output confirmed that there are no missing values in any of the dataset's features.

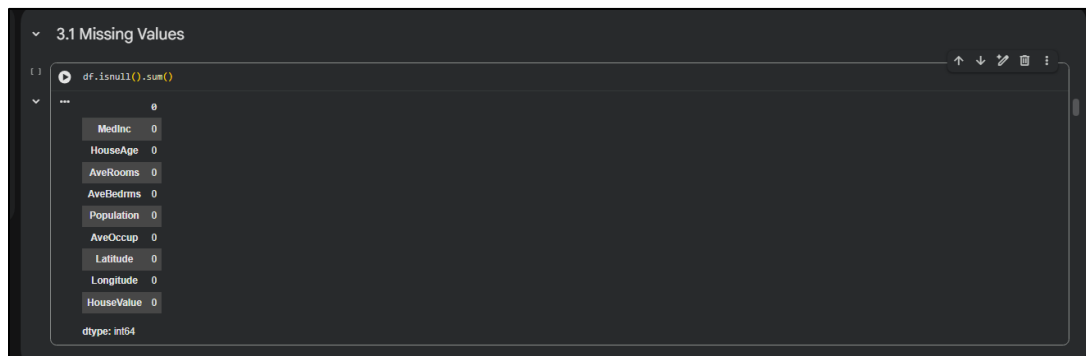


Figure1: Output of `df.isnull().sum` showing zero missing values in all the columns.

- **Statistical Auditing:** Ran `df.describe()` to analyze statistical means, standard deviations, and feature ranges.

4. Data Visualization

- **Histograms:** Generated distribution curves for all 9 attributes to inspect data skewness and variations.
- **Correlation Heatmap:** Constructed a Seaborn heatmap. The matrix proved **Median Income (MedInc)** holds the strongest positive linear correlation (~ 0.69) with **HouseValue**, identifying it as the main predictive driver.

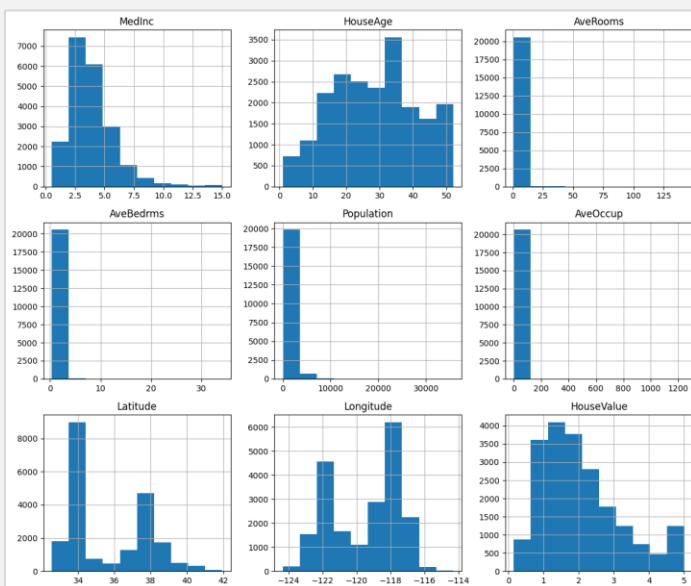


Figure2: Feature histograms and target variable distribution for the California Housing Dataset

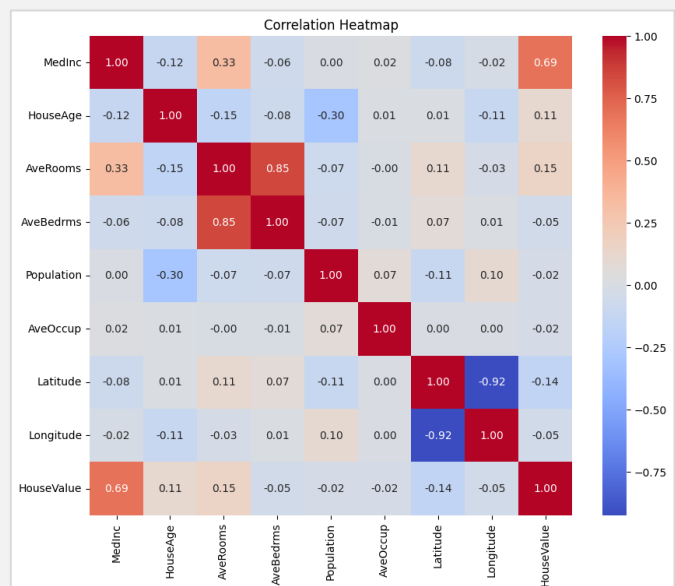


Figure3: Pearson correlation coefficient matrix for input features and target value.

5. Split Features and Target

The matrix space was isolated into independent variables (X) and the dependent target (y). The dataset was divided into an **80% training / 20% testing split** using random_state=42 to guarantee strict evaluation reproducibility and prevent data leakage.

6. Feature Scaling

Because features vary widely in scale (e.g., Population spans thousands while MedInc stays under 15), Scikit-learn’s StandardScaler was applied to shift feature distributions to a mean of 0 and standard deviation of 1. The test data was transformed strictly using the training scale parameters.

7. Train the Linear Regression Model

An Ordinary Least Squares (OLS) LinearRegression model was initialized and fitted directly onto X_train_scaled and y_train, optimizing the feature coefficients and the baseline intercept.

8. Model Evaluation

Evaluated against the unseen 20% testing data, the baseline model achieved:

Mean Absolute Error (MAE):	0.5332
Root Mean Squared Error (RMSE):	0.7456
R ² Score (Coefficient of Determination):	0.5758 (Captures 57.6% of housing price variance).

Diagnostic Visualizations:

- Actual vs. Predicted Scatter Plot:** Displays a clear diagonal positive trendline while revealing the dataset's native \$500,001 pricing cap.
- Residual Plot:** Plots error variances around the \$y=0\$ horizontal baseline, highlighting minor heteroscedasticity at top-tier valuation intervals.

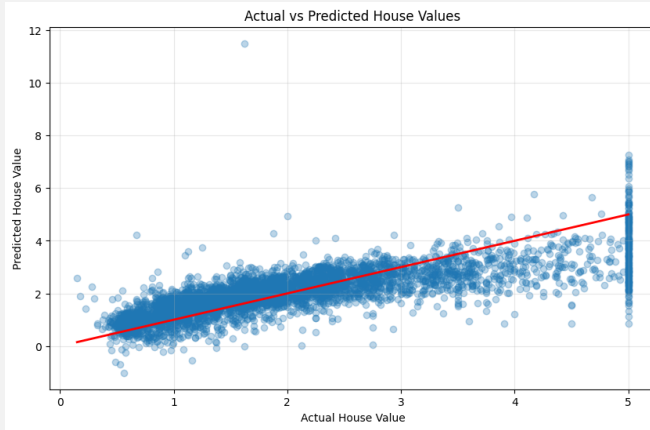


Figure3: Actual vs. Predicted house values showing model alignment along the diagonal baseline.

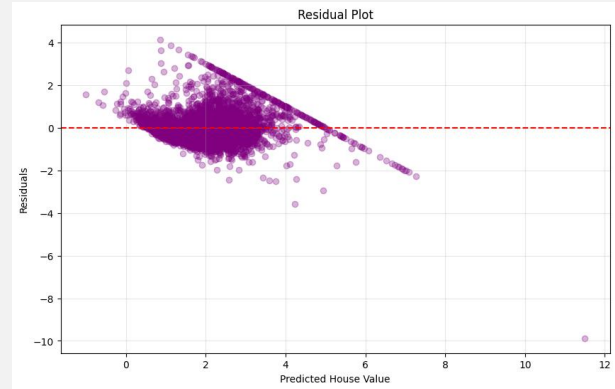


Figure3: Residual plot illustrating prediction errors centered around zero with mild heteroscedasticity.

9. Save the Trained Model

The finalized model state was successfully serialized to disk as `house_price_model.pkl` using Python's pickle module, making it ready for Streamlit dashboard integration.

10. Future Improvements

- **Non-Linear Models:** Benchmarking the baseline linear framework against tree-based ensembles like **Random Forest** or **XGBoost** to capture complex spatial interactions.
- **Feature Engineering:** Creating interaction ratios (e.g., bedrooms-to-rooms ratio) to reduce collinearity.

11. Conclusion

This project successfully implemented an end-to-end Machine Learning pipeline for California House Price Prediction.

The Linear Regression model achieved an **R² Score of 0.5758**, demonstrating satisfactory predictive performance for a baseline regression model. This project strengthened practical understanding of supervised learning, regression analysis, model evaluation, and deployment, while providing a strong foundation for implementing more advanced machine learning models in future work.